

NSF NATIONAL INNOVATION CONVENING



Innovation Acceleration Resource Guide

Accelerating Innovation Discovery and Commercialization Translation:

A National Dialogue with the Innovation Community

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Purpose of This Guide

The National Innovation Convening brings together leaders and stakeholders from across the U.S. innovation ecosystem to explore how the nation can more effectively move promising discoveries from **research to translation, commercialization, adoption, and societal impact**.

This resource guide provides a common intellectual and policy foundation to inform the discussions. It combines:

Part I — Evidence Base: Selected Peer-Reviewed Research

This section highlights research addressing innovation, artificial intelligence, commercialization, technology transfer, innovation ecosystems, and the barriers that often separate discovery from impact.

Part II — National Innovation Policy & Strategy Resources

Selected federal and National Academies resources illustrating current strategies, programs, and policy approaches designed to strengthen U.S. innovation capacity.

Part III — Pre-Convening Reflection Questions

This section provides questions designed to help participants arrive prepared to contribute ideas and actionable recommendations to the **National Innovation Roadmap**.

These resources are **recommended rather than required reading**. Participants are encouraged to review the materials most relevant to your expertise and experience.

PART I

EVIDENCE BASE — SELECTED PEER-REVIEWED RESEARCH

1. Building the Capacity to Innovate

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1), 128–152.

Why it matters

This landmark article introduced the concept of **absorptive capacity**—the ability of an organization to recognize valuable external knowledge, assimilate it, and apply it commercially. The authors argue that prior knowledge and organizational learning capacity strongly influence innovation performance.

For the NIC, the concept raises an important question: producing excellent research is not enough. Organizations, regions, and innovation ecosystems must also possess the ability to **identify, absorb, translate, and deploy knowledge**.

Consider:

What capabilities do universities, companies, federal agencies, and regions need in order to recognize and rapidly act on promising discoveries?

DOI: 10.2307/2393553

2. Open Innovation and Cross-Sector Collaboration

Bogers, M., Chesbrough, H., & Moedas, C. (2018). Open innovation: Research, practices, and policies. *California Management Review*, 60(2), 5–16.

Why it matters

Innovation increasingly occurs across organizational boundaries rather than entirely within individual institutions. This article examines open innovation at the intersection of research, organizational practice, and public policy and discusses opportunities and challenges created by external knowledge flows and collaboration.

This framework is highly relevant to the NIC, where participants represent universities, companies, investors, entrepreneurs, government, technology-transfer organizations, and other parts of the innovation system.

Consider:

What institutional or policy barriers make it unnecessarily difficult for knowledge, technologies, talent, and capital to move across sectors?

DOI: 10.1177/0008125617745086

3. From Discovery to Transformative Innovation

Pettigrew, R. I., & Cooke, J. P. (2022). At the nexus of science, engineering, and medicine: Pasteur's quadrant reconsidered. *PNAS Nexus*, 1(3), pgac092.

Why it matters

The authors reconsider the traditional distinction between basic and applied research and emphasize the importance of convergence, interdisciplinary exchange, and movement among research approaches in producing transformative innovation.

The article encourages a more dynamic understanding of the pathway from discovery to impact.

Consider:

Should federal research systems more intentionally support movement between basic research, use-inspired research, application, prototyping, and commercialization?

DOI: 10.1093/pnasnexus/pgac092

Artificial Intelligence and Innovation

4. Generative AI and Professional Productivity

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654), 187–192.

Why it matters

In a preregistered experiment involving 453 college-educated professionals, researchers found that access to generative AI significantly increased productivity: average completion time declined by approximately 40%, while output quality increased by approximately 18%.

The findings suggest that AI can significantly alter how knowledge work is performed.

Consider:

If similar productivity improvements can occur in research, engineering, entrepreneurship, technology transfer, and commercialization, where could AI most significantly shorten the innovation cycle?

DOI: 10.1126/science.adh2586

5. Generative AI, Expertise, and Knowledge Diffusion

Brynjolfsson, E., Li, D., & Raymond, L. (2025). Generative AI at work. *The Quarterly Journal of Economics*, 140(2), 889–942.

Why it matters

This large-scale study examined the introduction of a generative AI assistant among 5,172 customer-support agents. The research found significant productivity benefits and provides important evidence regarding how AI can transfer knowledge and capability across a workforce.

For innovation systems, this raises the possibility that AI could help broaden access to specialized knowledge that historically resided within a relatively small number of experts.

Consider:

Could AI democratize access to expertise in areas such as entrepreneurship, intellectual property, regulatory navigation, engineering design, market assessment, and commercialization?

DOI: 10.1093/qje/qjae044

6. Understanding the Limits of AI

Dell'Acqua, F., et al. (2026). Navigating the jagged technological frontier: Field experimental evidence of the effects of artificial intelligence on knowledge worker productivity and quality. *Organization Science*, 37(2), 403–423.

Why it matters

A preregistered experiment involving 758 knowledge workers demonstrated that AI significantly improved performance on tasks falling within its capabilities, but the researchers also identified an important “**jagged technological frontier**”: AI is highly capable on some tasks while remaining unreliable on seemingly similar ones.

This finding is important for avoiding an overly simplistic assumption that AI automatically improves every aspect of innovation.

Consider:

Where should innovators rely heavily on AI, and where must human expertise, judgment, validation, and oversight remain dominant?

DOI: 10.1287/orsc.2025.21838

Commercialization and the “Valley of Death”

7. Open Innovation and the Commercialization Gap

Stefan, I. (2022). Does open innovation enable or hinder crossing the valley of death? *International Journal of Innovation Management*, 26(9).

Why it matters

The article examines whether open innovation can help organizations move technologies from R&D to commercialization. It finds that collaboration can help bridge the commercialization gap through mechanisms such as collaborative platforms and inter-industry partnerships, but collaboration can also introduce challenges including conflicting interests and opportunistic behavior.

Consider:

What forms of collaboration are most effective at moving technologies across the valley of death—and what makes some partnerships fail?

DOI: 10.1142/S1363919622400254

8. Understanding the Challenges Facing Technology Startups

Gbadegeshin, S. A., et al. (2022). Overcoming the valley of death: A new model for high technology startups. *Sustainable Futures*, 4, 100077.

Why it matters

Drawing on a review of 128 scholarly sources and empirical data from 30 technology startups, this study develops a framework for understanding the challenges that emerging technology ventures encounter during their earliest stages.

The barriers extend beyond financing and include organizational capacity, market readiness, technology maturity, partnerships, and commercialization strategy.

Consider:

Are today's innovation programs addressing the entire commercialization journey—or primarily the funding gap?

DOI: 10.1016/j.sftr.2022.100077

9. Policy Barriers to Translation

Meslin, E. M., Blasimme, A., & Cambon-Thomsen, A. (2013). Mapping the translational science policy “valley of death.” *Clinical and Translational Medicine*, 2, 14.

Why it matters

This article broadens the valley-of-death concept by demonstrating that policy itself can create barriers to translation. Regulatory, ethical, institutional, and governance environments can slow progress just as significantly as scientific or financial limitations.

Consider:

What policies or administrative practices intended to support or protect innovation may unintentionally slow it?

DOI: 10.1186/2001-1326-2-14

Universities, Technology Transfer, and Entrepreneurship

10. Technology Transfer Offices and Commercialization

Siegel, D. S., Veugelers, R., & Wright, M. (2007). Technology transfer offices and commercialization of university intellectual property: Performance and policy implications. *Oxford Review of Economic Policy*, 23(4), 640–660.

Why it matters

The authors examine university technology-transfer performance and conclude that universities and regions need coherent, feasible commercialization strategies rather than treating patenting and licensing as isolated activities.

Consider:

Should the next generation of technology-transfer organizations be measured primarily by patents and licenses—or by broader measures of economic and societal impact?

DOI: 10.1093/oxrep/grm036

11. Academic Entrepreneurship After Bayh-Dole

Grimaldi, R., Kenney, M., Siegel, D. S., & Wright, M. (2011). 30 years after Bayh-Dole: Reassessing academic entrepreneurship. *Research Policy*, 40(8), 1045–1057.

Why it matters

This article examines the evolution of university commercialization and academic entrepreneurship since enactment of the Bayh-Dole Act and considers the institutional, managerial, and policy structures influencing university-based innovation.

Consider:

Are university incentives, promotion systems, intellectual-property practices, and organizational structures adequately designed to support translation and entrepreneurship?

DOI: 10.1016/j.respol.2011.04.005

Innovation Ecosystems

12. Measuring Entrepreneurial Ecosystems

Johnson, E. E., Hemmatian, I., Lanahan, L., & Joshi, A. M. (2022). A framework and databases for measuring entrepreneurial ecosystems. *Research Policy*, 51(2), 104398.

Why it matters

This work places relationships among entrepreneurs, firms, government organizations, accelerators, and research institutions at the center of entrepreneurial ecosystem performance. It also highlights multiple roles government can play, including catalyst, coordinator, certifier, and customer.

Consider:

What indicators should the United States use to determine whether an innovation ecosystem is truly healthy and productive?

DOI: 10.1016/j.respol.2021.104398

PART II

NATIONAL INNOVATION POLICY & STRATEGY RESOURCES

1. NSF FY 2026–2030 Strategic Plan

The current NSF Strategic Plan identifies transformative research and innovation, U.S. technological leadership, STEM talent, partnerships, and measurable impact as major agency priorities. It also asks an especially relevant question for the NIC: how NSF can better measure the outcomes and impact of its investments in U.S. research and innovation.

NIC relevance:

This provides an important overarching framework for considering how federal research investments translate into economic growth, national security, technological leadership, and societal benefit.

Roadmap question:

How should the United States define and measure the impact of publicly supported innovation?

2. NSF Directorate for Technology, Innovation and Partnerships — TIP

NSF TIP focuses explicitly on **use-inspired and translational research** and organizes its work around accelerating critical and emerging technologies, expanding the geography of innovation, and building a competition-ready workforce. Its portfolio includes Accelerating Research Translation, I-Corps, Regional Innovation Engines, Translation to Practice, America's Seed Fund, technology accelerators, and other programs.

NIC relevance:

TIP provides a contemporary federal model for connecting research, technology translation, entrepreneurship, regional development, workforce, and industry partnerships.

Roadmap question:

What gaps remain between excellent federal innovation programs and the ability to move discoveries rapidly and routinely into use?

3. NSF Innovation Corps — I-Corps™

NSF I-Corps provides immersive entrepreneurial education that helps scientists and engineers move beyond the laboratory, understand customer and market needs, and accelerate the transformation of invention into impact. NSF identifies technology translation, entrepreneurial workforce development, economic impact, and innovation ecosystems among the program's central objectives.

NIC relevance:

I-Corps demonstrates the importance of combining technology development with customer discovery, mentorship, market understanding, and entrepreneurial capability.

Roadmap question:

Which elements of the I-Corps model should be expanded, adapted, or connected more directly with later stages of commercialization?

4. NSF Translation to Practice

NSF's Translation to Practice initiative supports use-inspired research, translational activity, partnerships, commercial applications, market learning, and industry adoption. NSF explicitly recognizes that translation from scientific discovery to societal impact often follows nonlinear pathways.

NIC relevance:

The program directly addresses one of the central questions of the Convening: how the national innovation system can shorten the distance between research and real-world application.

Roadmap question:

What support is most commonly missing between proof-of-concept and adoption?

5. NSF Regional Innovation Engines

NSF Engines are designed to create sustainable, place-based innovation ecosystems built around cross-sector coalitions involving academia, industry, government, nonprofits, communities, entrepreneurs, and workforce organizations. Their core functions include use-inspired R&D, translation of innovation results to society, and workforce development.

NIC relevance:

The Engines model encourages participants to consider innovation as an **ecosystem-level challenge**, rather than simply the responsibility of individual researchers or startups.

Roadmap question:

What makes an innovation ecosystem self-sustaining after initial federal investment ends?

6. America's AI Action Plan

The federal AI strategy emphasizes acceleration of AI innovation, development of supporting infrastructure, workforce and scientific capacity, and strengthening U.S. leadership in artificial intelligence.

NIC relevance:

AI should be considered not only as an industry or emerging technology, but also as a potential **horizontal innovation accelerator** capable of affecting discovery, design, modeling, experimentation, decision-making, manufacturing, commercialization, and workforce development.

Roadmap question:

Where can AI responsibly remove time, cost, and friction from today's innovation process?

7. National Academies — Strategic Innovation and Commercialization

National Academies of Sciences, Engineering, and Medicine. *Strategic Innovation and Commercialization: Supporting IP and Tech Transfer to Advance U.S. Research Competitiveness.*

This National Academies resource examines intellectual property, technology transfer, commercialization, partnerships, and research competitiveness.

NIC relevance:

It invites consideration of whether today's technology-transfer structures adequately support an innovation system increasingly characterized by interdisciplinary research, rapid technological change, global competition, and cross-sector partnerships.

Roadmap question:

Which parts of the current technology-transfer system enable commercialization, and which create unnecessary friction?

8. NIST — Federal Technology Transfer and Laboratory-to-Market Activities

NIST plays an important coordinating role in federal technology transfer and tracks mechanisms including patents, licensing, Cooperative Research and Development Agreements, and other approaches for transferring federally developed technologies toward commercial or societal use.

NIC relevance:

Federal laboratories represent an enormous source of research expertise and intellectual property,

yet their integration with universities, startups, investors, and regional ecosystems can vary considerably.

Roadmap question:

How can federal laboratories participate more seamlessly in America's broader innovation ecosystem?

9. CHIPS for America — Research and Development Programs

CHIPS R&D provides an instructive model of a mission-driven innovation strategy intended to connect invention, development, prototyping, deployment, workforce development, and commercial scale. NIST describes three goals: building domestic capacity to invent and deploy semiconductor technologies, accelerating ideas to market, and developing the necessary semiconductor workforce.

NIC relevance:

The CHIPS model provides an opportunity to examine whether integrated approaches involving shared infrastructure, prototyping, workforce investment, industry participation, and commercialization could be applied to other critical technology areas.

Roadmap question:

What aspects of mission-driven models such as CHIPS could accelerate commercialization in other industries?

PART III

PRE-CONVENING REFLECTION QUESTIONS

Participants are encouraged to reflect on the following questions before arriving at the National Innovation Convening.

1. Identify the Barrier

What is the single greatest barrier you encounter—or observe—in moving promising discoveries toward implementation or commercialization?

Consider barriers involving:

- Funding
- Technology readiness
- Market validation
- Regulatory processes
- Intellectual property
- University policies
- Technology-transfer processes
- Access to industry partners
- Access to investors
- Entrepreneurial expertise
- Talent and workforce
- Infrastructure and prototyping
- Risk tolerance
- Organizational culture
- Federal policies or administrative requirements

2. Identify the Intervention

If you could change one policy, practice, incentive, program, or institutional structure to remove that barrier, what would you change?

Participants should be prepared to move beyond describing problems and suggest **actionable interventions**.

3. Consider the Innovation Journey

Think about the pathway:

**DISCOVERY → TRANSLATION → VALIDATION →
COMMERCIALIZATION → SCALE → IMPACT**

Where does the innovation process most frequently slow down or fail?

What resources, partnerships, capabilities, or policies are missing at that stage?

4. Consider Artificial Intelligence

Where could AI have the greatest positive impact on the innovation lifecycle?

Consider:

- Scientific discovery
- Literature and knowledge synthesis
- Experimental design
- Engineering design
- Simulation and digital twins
- Prototyping
- Customer discovery
- Market analysis
- Intellectual-property analysis
- Regulatory preparation
- Manufacturing
- Supply-chain planning
- Entrepreneurship
- Investor readiness
- Technology-transfer processes

Also consider:

Where should AI not replace human judgment?

5. Consider the Innovation Ecosystem

Which relationships must become stronger for innovation to move faster?

Examples may include:

University ↔ Industry
University ↔ Investors
Researchers ↔ Entrepreneurs
Federal Agencies ↔ Industry
Federal Laboratories ↔ Universities
Startups ↔ Large Corporations

Research Institutions ↔ Communities
Universities ↔ Regional Economic Development Organizations

What currently prevents those relationships from working effectively?

6. Consider Incentives

Are researchers, universities, companies, federal laboratories, investors, and other innovation stakeholders rewarded for the behaviors necessary to accelerate innovation?

If not:

What incentives need to change?

7. Consider Access to Innovation

Who currently has access to America's innovation ecosystem—and who does not?

Consider geographic location, institutional size, research infrastructure, access to capital, access to networks, workforce availability, and participation by emerging innovation regions.

How can the nation's innovation capacity be expanded beyond the institutions and regions that have historically dominated technology commercialization?

8. Consider Scale

The United States has many successful innovation programs.

The larger question is whether successful models can be connected and scaled.

Which existing program, model, or practice should be expanded nationally?

What would be required to do so?

THE NATIONAL INNOVATION ROADMAP CHALLENGE

Throughout the Convening, participants will be asked to move from:

Observation → Barrier Identification → Root Cause → Intervention → Action

The goal is not simply to identify what is wrong with the innovation system.

The goal is to identify **what can be changed**.

As you prepare for the National Innovation Convening, consider:

If the United States could make only three major changes during the next five years to dramatically accelerate innovation discovery and commercialization, what should those changes be?

Your insights, experience, and recommendations will help inform the development of the:

NATIONAL INNOVATION ROADMAP

From Discovery to Impact

Accelerating American Innovation through Research, Technology, Talent, Partnerships, and Translation